SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	SANTHOSH G	**Reg. No.:**	192521035
Date:	20-10-2026	**Duration:**	60 minutes

Code of Conduct

I SANTHOSH G-192521035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SANTHOSH G

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Agile Sprint Planning and User Story Workshop

1. Project Vision and Stakeholders

1.1 Background

Universities today manage thousands of students, hundreds of courses and many departments at the same time. In most institutions, admissions, course registration, attendance, examinations and academic records are handled in separate tools or on paper. This leads to duplicated and inconsistent data, late reports, poor communication between students, faculty and administrators, and – most importantly – late discovery of students who are struggling. By the time a failing grade appears, the opportunity for support has already passed.

The assigned project is to build a University Academic Management and Student Success Platform: one centralised system that manages the whole academic lifecycle and continuously checks attendance, marks and enrolment against configurable academic policies, so that at-risk students can be identified and helped early. This document applies Agile Scrum practices to plan the first stages of that product.

1.2 Project Vision

"For universities that struggle with scattered academic data and late intervention, the University Academic Management and Student Success Platform is a secure, scalable web platform that manages admissions, registration, attendance, examinations and records in one place and uses configurable policies to detect at-risk students early. Unlike disconnected spreadsheets and standalone tools, it connects every academic activity to student success."

The vision is aligned with the problem statement in three ways. It is centralised (one source of truth for academic data), proactive (monitoring and alerts instead of after-the-fact reporting) and scalable (multiple departments, programmes and thousands of students with high availability and security).

1.3 Project Objectives

- Maintain a single, accurate and secure academic record for every student, course and faculty member.
- Allow online course registration with automatic checks for seat availability, prerequisites and timetable clashes.
- Monitor attendance and academic performance continuously against configurable institutional policies.
- Manage the full examination process, from scheduling to result publication and revaluation.
- Identify academically at-risk students and notify students, faculty and advisors in time for intervention.

- Provide dashboards and reports that support evidence-based academic planning.
- Reduce manual administrative work and improve transparency, security, reliability and availability.

1.4 Measurable Success Indicators

To make the vision testable, the Product Owner and the team agree on the following indicators for the first releases:

- A student can complete course registration online in under 10 minutes, with no visit to the office.
- Attendance entered by a faculty member is reflected in each student's record within 5 seconds.
- A student whose attendance or marks cross a policy threshold is flagged within one day.
- At least 80% of faculty use the platform for attendance within the first semester after release.
- The platform achieves 99.9% availability during registration and examination periods.

1.5 Key Stakeholders

Stakeholders are all the people and groups who use, fund, manage or are affected by the platform. They are identified below with their responsibilities and what they expect from the system.

- **Students.** The primary users. They register for courses, view attendance, marks and results, and receive alerts. They expect transparent, real-time information about their progress, easy registration, and early feedback when they are falling behind.
- **Faculty members.** They mark attendance, set assessments, enter marks and monitor their classes. They need fast data entry that saves time and a clear list of students who need attention.
- **Academic advisors and mentors.** They approve registrations, review at-risk students and record interventions. They need early warnings and one consolidated view of each student's attendance, marks and history.
- **Department heads (HoD).** They plan course offerings, allocate faculty and review department performance. They expect dashboards and trend reports to support decisions.
- **Administrators and Registrar.** They manage student records, programmes, academic policies and user accounts. They need accurate data, automation of repetitive work and a complete audit trail.
- **Admissions officers and applicants.** Applicants submit applications and track status; officers verify documents and prepare merit lists. Both expect a fair, quick and trackable process.
- **Examination controller.** Responsible for scheduling examinations, hall tickets, result publication and revaluation. Needs a secure, confidential and error-free workflow.
- **Senior management (Dean / Vice-Chancellor).** They use institution-wide analytics for strategic planning, quality assurance and accreditation reports.
- **Scrum team (Product Owner, Scrum Master, developers, tester, designer).** The Product Owner owns and orders the backlog, the Scrum Master facilitates the process, and the development team builds and tests the product. They need clear requirements and realistic sprint commitments.

- **IT operations.** They deploy, secure, monitor and support the platform, and expect a maintainable and monitorable system with high availability.

2. Epics and User Stories

2.1 Approach

The requirements in the problem statement were first grouped into large bodies of work called epics, each corresponding to a business capability. Each epic was then split into user stories written from the point of view of a real user in the format "As a <user>, I want <feature>, so that <benefit>." Stories were checked against the INVEST guidelines: they should be Independent, Negotiable, Valuable, Estimable, Small enough for one sprint, and Testable.

Epic E1 – User and Access Management

Covers secure authentication, role-based access control and account recovery. Every other module depends on knowing who the user is and what they may do.

- **US01:** As a user (student, faculty or administrator), I want to log in securely with role-based access, so that I can use only the features and data relevant to my role.
- **US08:** As a user, I want to reset my password and use multi-factor login, so that my account remains secure.

Epic E2 – Student Academic Records

Covers creation and maintenance of the permanent academic record and official documents such as transcripts.

- **US02:** As an administrator, I want to create and maintain student profiles, so that every student has one accurate and complete academic record.
- **US09:** As a student, I want to download my official transcript, so that I can share my academic record without waiting for the office.

Epic E3 – Admissions

Covers online applications, verification and rule-based selection of candidates.

- **US10:** As an applicant, I want to apply online and track my application status, so that I know my admission progress.
- **US11:** As an admissions officer, I want a rule-based merit list to be generated, so that selection is fair and fast.

Epic E4 – Programme and Course Management

Covers the academic catalogue: departments, programmes, courses, credits, prerequisites and seat limits.

- **US03:** As an administrator, I want to define programmes and courses with credits, prerequisites and seat limits, so that a correct academic catalogue is available for registration.

Epic E5 – Course Registration

Covers online enrolment with rule checks, and advisor approval of changes.

- **US04:** As a student, I want to register for courses online, so that I can enrol without visiting the office and without timetable clashes.
- **US12:** As an academic advisor, I want to approve or reject registration and add/drop requests, so that students follow suitable study plans.

Epic E6 – Attendance Management

Covers recording and viewing attendance, setting policy thresholds and warning students about shortages.

- **US05:** As a faculty member, I want to mark attendance for my class sessions online, so that attendance records are accurate and saved instantly.
- **US06:** As a student, I want to view my course-wise attendance, so that I know whether I am meeting the attendance requirement.
- **US07:** As an administrator, I want to configure attendance thresholds, so that policy changes can be applied without code changes.
- **US13:** As a student, I want to be alerted when my attendance falls below the threshold, so that I can recover in time.

Epic E7 – Assessment and Examination

Covers marks entry, examination scheduling, results, GPA calculation and revaluation.

- **US14:** As a faculty member, I want to enter assessment marks online, so that grading is accurate and quick.
- **US15:** As an examination controller, I want to schedule exams and generate hall tickets, so that exams are conducted without clashes.
- **US16:** As a student, I want to view published results and my GPA, so that I can track my academic performance.
- **US17:** As a student, I want to request revaluation of a result, so that I can contest possible errors.

Epic E8 – Student Success and Early Warning

The core differentiator of the product: policy-based identification of at-risk students and tracking of the support given to them.

- **US18:** As an academic advisor, I want the system to flag at-risk students using configured policies, so that I can intervene early.
- **US19:** As an academic advisor, I want to record and track interventions for at-risk students, so that follow-up is documented and effective.

Epic E9 – Notifications and Communication

Covers email, SMS and in-app alerts used by all other epics.

- **US20:** As a user, I want to receive notifications by email, SMS and in-app messages, so that I am informed of important academic events.

Epic E10 – Reports, Analytics and Faculty Activity

Covers dashboards, institutional reports and faculty workload management for decision-making.

- **US21:** As a department head, I want a dashboard of attendance, pass percentage and at-risk counts, so that I can make timely decisions.
- **US22:** As a dean, I want institutional reports exported as PDF or Excel, so that planning and accreditation are supported.
- **US23:** As a department head, I want to allocate courses and view faculty workload, so that teaching load is balanced.

In total, 10 epics and 23 user stories were defined. Together they cover every objective and expected outcome in the problem statement: records, registration, attendance and performance monitoring, examinations, notifications, reports, at-risk identification, planning support and scalability.

3. Acceptance Criteria (Given–When–Then)

Acceptance criteria describe exactly when a story is considered complete. They are written in the Given–When–Then format, are specific and measurable, and can be directly converted into test cases. They are given here for the seven stories selected for Sprint 1 (US01 to US07).

US01 – Secure login with role-based access

Scenario 1: Valid login

Given A registered user with valid credentials

When they submit the login form

Then they are redirected to their role-specific dashboard within 3 seconds.

Scenario 2: Account lockout

Given a user enters a wrong password

When they fail 5 consecutive times

Then the account is locked for 15 minutes and a generic error is shown that does not reveal which field was wrong.

Scenario 3: Role restriction

Given a student is logged in

When they try to open an administrator page

Then access is denied (HTTP 403) and the attempt is recorded in the log.

US02 – Create and maintain student profiles

Scenario 1: Create profile

Given an administrator is logged in

When they submit the student form with all mandatory fields

Then a unique student ID is generated, the record is saved and it appears in search within 2 seconds.

Scenario 2: Duplicate check

Given a student with the same email or ID number already exists

When the administrator submits the form

Then the system rejects it and names the duplicate field.

Scenario 3: Audit of changes

Given a student record exists

When the administrator edits any field

Then the change is stored with the user, timestamp, old value and new value.

US03 – Define programmes and courses

Scenario 1: Create course

Given an administrator is logged in

When they enter course code, title, credits (1–6), prerequisites and seat limit and save

Then the course appears in the catalogue and can be filtered by department and programme.

Scenario 2: Duplicate course code

Given a course with the same code already exists

When the administrator saves another course with that code

Then the save is rejected with a clear message.

US04 – Online course registration

Scenario 1: Successful registration

Given the registration window is open, seats are available and prerequisites are met

When the student registers for a course

Then enrolment is confirmed, the seat count reduces by 1 and a confirmation is shown within 3 seconds.

Scenario 2: Course full

Given a course has no seats left

When a student tries to register

Then registration is rejected with the message “Course full”.

Scenario 3: Prerequisite or clash

Given a prerequisite has not been passed or the timetable overlaps

When a student tries to register

Then registration is rejected and the exact reason is displayed.

Scenario 4: Window closed

Given the registration window has ended

When a student opens the registration page

Then the register option is disabled and the closing date is shown.

US05 – Faculty marks attendance

Scenario 1: Mark attendance

Given a faculty member is assigned to a course session

When they mark every student present or absent and submit

Then the record is saved and each student’s attendance percentage is updated within 5 seconds.

Scenario 2: Unauthorised faculty

Given a faculty member is not assigned to the course

When they try to open its attendance sheet

Then the sheet is not accessible.

Scenario 3: Edit window

Given attendance has already been submitted

When the faculty member edits it

Then within 24 hours the edit is saved with an audit entry; after 24 hours it requires administrator approval.

US06 – Student views attendance

Scenario 1: View attendance

Given a student is logged in

When they open the attendance page

Then course-wise classes held, classes attended and percentage are shown and match the faculty records exactly.

Scenario 2: Below threshold

Given a course percentage is below the configured threshold

When the student views the page

Then that course is highlighted with a warning label.

US07 – Configure attendance thresholds

Scenario 1: Set threshold

Given an administrator is logged in

When they enter a minimum attendance (for example 75%) for a programme and save

Then the policy is stored with an effective date and applied to calculations without any code change.

Scenario 2: Invalid value

Given an administrator is logged in

When they enter a value below 0 or above 100

Then the save is rejected with a validation message.

4. Product Backlog Prioritisation

4.1 Prioritisation Method

The Product Backlog is an ordered list of everything needed in the product. The order was decided using three factors agreed with the Product Owner:

- **Business value:** each story was scored from 1 (low) to 10 (high) according to how much it contributes to the vision, such as reducing manual work or enabling early intervention.
- **Customer need:** stories were classified using MoSCoW – Must have, Should have and Could have – based on what stakeholders cannot do without versus what is desirable.

- **Dependencies:** a story is never scheduled before the stories it technically relies on. For example, registration needs student records and the course catalogue, and shortage alerts need attendance, thresholds and notifications.

4.2 Ordered Product Backlog

Rank	ID	Story	Value (1-10)	MoSCoW	Depends on	Sprint
1	US01	Secure login with role-based access	10	Must	None	Sprint 1
2	US02	Create and maintain student profiles	9	Must	US01	Sprint 1
3	US03	Define programmes and courses	9	Must	US01	Sprint 1
4	US04	Online course registration	10	Must	US02, US03	Sprint 1
5	US05	Faculty marks attendance	9	Must	US02, US03	Sprint 1
6	US06	Student views attendance	8	Must	US05	Sprint 1
7	US07	Configure attendance thresholds	7	Must	US01	Sprint 1
8	US20	Notification service	8	Must	US01	Sprint 2
9	US13	Attendance shortage alert	9	Must	US05, US07, US20	Sprint 2
10	US14	Faculty enters assessment marks	9	Must	US03, US04	Sprint 2
11	US08	Password reset and multi-factor login	6	Should	US01	Sprint 2
12	US12	Advisor approval and add/drop	7	Should	US04	Sprint 2
13	US16	Results publication and GPA	9	Must	US14	Sprint 3
14	US18	At-risk student identification	10	Must	US05, US14, US07	Sprint 3
15	US19	Intervention tracking	8	Should	US18	Sprint 3
16	US10	Online admission application	8	Should	US01	Sprint 4
17	US15	Exam scheduling and hall tickets	7	Should	US03, US04	Sprint 4
18	US21	Department dashboard	8	Should	US18	Sprint 4
19	US09	Transcript generation	7	Should	US16	Sprint 5
20	US11	Merit list generation	6	Should	US10	Sprint 5
21	US22	Institutional reports export	7	Should	US21	Sprint 5

Rank	ID	Story	Value (1-10)	MoSCoW	Depends on	Sprint
22	US17	Revaluation request	5	Could	US16	Sprint 6
23	US23	Faculty workload management	5	Could	US03	Sprint 6

4.3 Release Roadmap and Rationale

Sprint 1 – Foundation and first value. Login, student records and the course catalogue are built first because everything else depends on them. Registration and attendance are included because they are the highest-value daily processes and the data source for early warning later.

Sprint 2 – Communication and marks. The notification service, attendance shortage alerts and marks entry are added, together with account security and advisor approval.

Sprint 3 – Student success. Results and GPA are published and the at-risk identification engine and intervention tracking are delivered. This is the main differentiator of the product and needs attendance and marks data from earlier sprints.

Sprint 4 to 6 – Completion. Admissions, exam scheduling, dashboards, transcripts, merit lists, institutional reports, revaluation and faculty workload complete the platform. Could-have items are kept last so they can be dropped if time is short.

The backlog is a living document. It will be refined after every sprint review as stakeholders give feedback and policies change.

5. Sprint 1 Backlog and Task Breakdown

5.1 Sprint Details

- **Sprint:** Sprint 1, lasting 2 weeks (10 working days).
- **Team:** 5 members – 3 developers, 1 tester and 1 UI/UX designer – along with the Scrum Master and Product Owner.
- **Capacity:** 5 members × 10 days × 6 productive hours = 300 hours. About 30 hours are reserved for Scrum events (planning, daily stand-ups, review, retrospective), leaving 270 hours available for tasks.
- **Selected stories:** US01, US02, US03, US04, US05, US06 and US07 – the top seven items of the product backlog.

5.2 Why These Stories Were Selected

These stories were chosen because they have no dependency outside the sprint and together form a usable vertical slice. An administrator can create students and courses (US02, US03), users can log in securely (US01), students can register (US04), faculty can record attendance (US05), and students and administrators can see and configure attendance rules (US06, US07). Stories that need notifications or marks, such as shortage alerts and results, were deliberately left for Sprint 2 to keep the commitment realistic.

5.3 Task Breakdown

Each story is divided into small tasks that one team member can finish in about one to two days. Estimates are in hours.

Sprint setup (shared)

- T0.1 Create repository, CI/CD pipeline and dev/test environments – 12 h
- T0.2 Set up UI template, coding standards and project structure – 6 h

US01 – Secure login

- T1.1 Design user and role database schema – 6 h
- T1.2 Build login API with hashed passwords and token session – 10 h
- T1.3 Implement role-based access middleware – 8 h
- T1.4 Build login page and role dashboards – 10 h
- T1.5 Write unit and security tests (lockout, 403) – 6 h

US02 – Student profiles

- T2.1 Design student profile schema – 4 h
- T2.2 Build create/edit/search API with validation and duplicate check – 10 h
- T2.3 Build student form and search UI – 10 h
- T2.4 Implement audit logging for changes – 6 h
- T2.5 Write tests – 6 h

US03 – Programmes and courses

- T3.1 Design schema for programmes, courses and prerequisites – 5 h
- T3.2 Build course CRUD API – 10 h
- T3.3 Build administrator course management UI – 9 h
- T3.4 Add validations and tests – 6 h

US04 – Course registration

- T4.1 Design enrolment schema and seat counter – 5 h
- T4.2 Build rule checks for seats, prerequisites and timetable clash – 14 h
- T4.3 Build registration API with transaction locking – 10 h
- T4.4 Build registration UI with window open/closed state – 10 h
- T4.5 Integration and concurrency tests – 10 h

US05 – Mark attendance

- T5.1 Design session and attendance schema – 5 h
- T5.2 Build attendance marking API – 8 h
- T5.3 Build class-list attendance UI – 10 h
- T5.4 Implement 24-hour edit rule with audit entry – 6 h

- T5.5 Write tests – 6 h

US06 – View attendance

- T6.1 Build attendance percentage calculation service – 6 h
- T6.2 Build student attendance page – 8 h
- T6.3 Add below-threshold highlighting – 3 h
- T6.4 Write tests – 4 h

US07 – Attendance thresholds

- T7.1 Create policy table and API – 6 h
- T7.2 Build administrator policy UI – 6 h
- T7.3 Connect threshold to calculation service – 4 h
- T7.4 Write tests – 3 h

The total planned effort is 248 hours (setup 18, US01 40, US02 36, US03 30, US04 49, US05 35, US06 21, US07 19), which is within the 270-hour capacity and leaves a buffer of about 8% for unexpected work.

6. Story Point Estimation

6.1 Technique Used

Estimation was done using Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13). Fibonacci numbers are used because the gaps widen as stories get larger, which reflects the fact that large stories are less certain and should not be estimated with false precision.

The process was as follows. The Product Owner explained a story, and the team asked questions. Every team member then privately selected a card, and all cards were revealed together to avoid influencing each other. Members with the highest and lowest votes explained their reasoning, and the team voted again until a consensus was reached.

Estimates are relative to a reference story. US06 (student views attendance) was chosen as the baseline and set to 3 points, because it is a simple read-only page built on data that already exists. Story points express complexity, effort and uncertainty together; they are not a measure of hours.

6.2 Estimates and Justification

US01 – 5 points (votes 3, 5, 5, 5, 8). Authentication, password hashing, lockout and role-based middleware are moderately complex. The approach is well known, but because it is security-sensitive and every later feature depends on it, it is larger than the reference story. The vote of 8 was lowered after the team agreed to use a standard authentication library.

US02 – 5 points (votes 3, 5, 5, 5, 5). This is standard create-read-update with validation, a duplicate check and audit logging. Uncertainty is low, but the number of fields and rules makes it larger than a simple page.

US03 – 5 points (votes 3, 5, 5, 5, 8). Basic course management is easy, but modelling prerequisites between courses and seat limits adds moderate complexity. The higher vote reflected uncertainty about prerequisite rules, which will be confirmed with the Product Owner.

US04 – 8 points (votes 5, 8, 8, 8, 13). This is the most complex story in the sprint. It combines three rule checks (seats, prerequisites, timetable clashes), control of the registration window and the need to handle many students registering at the same time, which requires transaction locking. It carries the highest risk and uncertainty, so it was estimated at 8 and the team agreed not to split it because its parts are not useful alone.

US05 – 5 points (votes 5, 5, 5, 3, 8). It involves linking faculty to sessions, a bulk marking screen for a full class, an edit window and auditing. The complexity is medium, and the team has done similar bulk-entry screens before.

US06 – 3 points (votes 3, 3, 2, 3, 3). The reference story: a read-only page, a percentage calculation and a warning highlight.

US07 – 3 points (votes 3, 2, 3, 3, 5). A simple configuration form and one table. The only added effort is connecting the threshold to the calculation service.

6.3 Summary and Velocity

Story	Points	Remark
US01, US02, US03	5 + 5 + 5 = 15	Foundation stories with moderate effort
US04	8	Highest-risk story
US05	5	Core faculty workflow
US06, US07	3 + 3 = 6	Small, quick stories
Total committed	34	Sprint 1 commitment

Because this is the team's first sprint, the expected velocity is assumed to be about 30 to 35 story points. A commitment of 34 points is realistic because the task-level estimate (248 hours) fits within the 270-hour capacity. After the sprint, the actual completed points will be used as the real velocity for planning Sprint 2.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, deliver a secure and working core of the platform in which administrators can create student records and the course catalogue, students can log in and register for courses online with automatic rule validation, and faculty can record attendance that students can view against a configurable threshold."

The goal is clear, achievable within two weeks and testable at the sprint review. It gives the team a single purpose when trade-offs are needed during the sprint.

7.2 Expected Deliverables

- **Secure access (US01).** A login page, role-specific dashboards for students, faculty and administrators, and access control that blocks unauthorised pages.
- **Student records (US02).** Working student profile management with search, duplicate detection and an audit trail of changes.
- **Course catalogue (US03).** Programmes and courses with credits, prerequisites and seat limits that administrators can create and update.
- **Course registration (US04).** Online registration that validates seat availability, prerequisites and timetable clashes and respects the registration window.
- **Attendance (US05, US06, US07).** Faculty attendance marking, a student attendance view with percentages and warnings, and administrator-configured thresholds.
- **Engineering assets.** A CI/CD pipeline, test and demo environments, automated tests and basic technical documentation.

7.3 Definition of Done

A story is counted as done only when all of the following are true:

- The code has been reviewed and merged to the main branch.
- All acceptance criteria have been verified by the tester and pass.
- Unit and integration tests pass in the CI pipeline.
- There are no open critical or high-severity defects.
- The feature is deployed to the test environment and demonstrated to the Product Owner.
- Documentation is updated.

7.4 Sprint Ceremonies

- **Sprint Planning:** On Day 1 the team confirms the sprint goal, selects stories and breaks them into tasks.
- **Daily Stand-up:** A 15-minute meeting every day to share progress, plans and blockers.
- **Backlog Refinement:** Mid-sprint, the team clarifies and estimates stories for Sprint 2 (US20, US13, US14, US08, US12).
- **Sprint Review:** On Day 10 the completed increment is demonstrated to stakeholders. The planned demo is: an administrator creates a course and a student; the student logs in and registers; a faculty member marks attendance; the student sees the percentage with a warning below the threshold.
- **Sprint Retrospective:** After the review, the team identifies what went well and what to improve in Sprint 2.

7.5 Risks and Mitigation

- **Complex registration logic:** US04 is complex and may overrun. The team will start it first, pair-program on the rule checks and, if needed, move only the UI polish to Sprint 2.

- **Unclear requirements:** Prerequisite rules may be unclear. They will be confirmed with the Product Owner on Day 1.
- **Environment or CI delays:** Environment problems could delay development. Task T0.1 will be completed in the first two days.

7.6 Conclusion

This workshop converted the problem statement into a clear vision, a set of epics and well-formed user stories with testable acceptance criteria, a prioritised backlog and a realistic first sprint. By delivering login, student and course records, registration and attendance in Sprint 1, the team creates the foundation for the platform's main goal – identifying at-risk students early – which will be delivered in Sprint 3.